

Ellectrify Grounding Blanket

20-second 3D product film, built and rendered entirely in Blender

This document is the full brief for building the video on the local PC (ASUS RTX 4080) with Claude Code driving Blender through Python. Hand the whole file to Claude Code together with the kickoff prompt in section 12. Everything is built from scripts so each shot can be rebuilt and re-rendered.

Item	Decision
Product	Ellectrify Grounding Blanket, US\$149.99
Core idea	It looks like an ordinary luxury quilt until it turns over and shows the silver. The film is built around that reveal, followed by an exploded view of the layers.
Length	20 seconds = 600 frames at 30 fps
Format	Vertical 9:16, 1080 x 1920, H.264 MP4 with AAC audio
Renderer	Cycles on the GPU (OptiX), OptiX denoiser
Blender version	Written for 4.2 LTS or newer. Claude Code must check the installed version first and adapt property names if it differs.
Look	Bright and calm: cream walls, soft daylight from the left, oak tones. Black appears only on the closing logo card.
Assembly	Six scenes rendered as image sequences, then cut together with text and sound in Blender's Video Sequencer.

1. Product facts

Only the first table is confirmed by the product page. Nothing in the second table may appear as a label or caption.

Confirmed	Source wording
Outer face	Plush white quilted cotton exterior
Inner face	Smooth metallic-grey conductive inner layer woven with fine silver threads
Connection	Snap-button grounding connection (the product photo shows a metal snap at a corner of the grey face)
Care	Machine washable
Queen size	60 x 80 in = 1.52 x 2.03 m. Use this size for every full-blanket shot.
Cord	White cord with a white snap cap at one end (seen on the brand's Instagram)

Not known	How the film handles it
Whether there is a padding layer between the two faces	Show a thin soft middle layer in the exploded view but give it no label.
Exact brand colours and typeface	Sample the yellow from the supplied logo file. Use the stand-in font in section 9 until the brand names its own.
Number and position of snaps	Model one snap, 40 mm in from one corner of the grey face.
Blanket thickness	Assume 15 mm. This is a visual choice, not a specification.

2. Rules that protect the pitch

- **No lettering on any 3D object.** The brand's own AI video misspelt its name on a mat. All text is added as clean type in the Sequencer.
- **No sparks, arcs or lightning effects.** The connection is shown as one slow, soft pulse of pale yellow light. The brand states that no electrical current touches the body.
- **No health or sleep claims beyond the brand's own published tagline.** Their compliance page says the products are wellness devices and not medical equipment.
- **No people or body parts.** It keeps the film product-only and avoids implying an effect on the body.
- **The logo is used only from the brand's own file.** Do not redraw or approximate it.

3. Files Saurav must supply before the build

File	Put it in	Used for
Ellectrify logo, SVG or transparent PNG (yellow on transparent)	assets/logo/	Shot 6 end card, and the sampled yellow for the pulse
Three blanket product photos saved from the product page	assets/ref/	Matching colour, quilting scale and snap shape
Instagram connector and cord image	assets/ref/	Modelling the cord and snap cap
Serif font files, regular and italic (TTF or OTF)	assets/fonts/	Labels and tagline
Music bed and four sound effects (section 10)	assets/audio/	Final mix

4. Project layout and how Claude Code runs Blender

```
ellectrify_blanket/  
  assets/          logo/  ref/  fonts/  audio/  
  scripts/         00_setup.py  10_assets.py  20_materials.py  
                   s1_folded.py  s2_unfurl.py  s3_exploded.py  
                   s4_snap.py    s5_bed.py    s6_logo.py  
                   80_render.py  90_edit.py  
  blend/           ellectrify_blanket.blend  (one file, one scene per shot)  
  cache/           cloth caches per shot  
  renders/         s1/ s2/ s3/ s4/ s5/ s6/  (PNG sequences)  
  previews/        one still per shot for sign-off  
  out/             ellectrify_blanket_v01.mp4
```

- Run each script headless: `blender -b blend/ellectrify_blanket.blend -P scripts/s3_exploded.py`. If a live Blender connection is available, the same scripts can be run through it so Saurav can watch the viewport.
- Every script must be safe to run twice: delete and rebuild its own objects by name, never duplicate them.
- Units are metres, Z is up, rotations are radians.
- Shared objects and materials live in a collection called LIB and are linked into each scene.

5. Global render settings

Setting	Value
Engine and device	Cycles, GPU Compute, OptiX. Enable the RTX 4080 in Preferences > System.
Resolution and rate	1080 x 1920 at 100%, 30 fps
Samples	Final: 192 with noise threshold 0.02. Preview stills: 48.
Denoise	OptiX, with albedo and normal passes
Light paths	Max bounces 8, glossy 4, transmission 4, clamp indirect 10
Colour	View transform AgX, look Base Contrast. Shot 6 uses Standard so the black stays true.
Motion blur	On, shutter 0.5. Off for shot 6.
Performance	Persistent data on
Output per shot	16-bit PNG sequence into renders/sN/
Final encode	H.264 MP4, constant rate factor 'High', slow preset, AAC 192 kbps

```
import bpy
p = bpy.context.preferences.addons['cycles'].preferences
p.compute_device_type = 'OPTIX'
p.get_devices()
for d in p.devices: d.use = (d.type == 'OPTIX')
for sc in bpy.data.scenes:
    sc.render.engine = 'CYCLES'; sc.cycles.device = 'GPU'
    sc.render.resolution_x, sc.render.resolution_y = 1080, 1920
    sc.render.fps = 30
    sc.cycles.samples = 192; sc.cycles.use_adaptive_sampling = True
    sc.cycles.adaptive_threshold = 0.02
    sc.cycles.use_denoising = True; sc.cycles.denoiser = 'OPTIX'
```

6. Assets to build

Name	Build notes
BLANKET_full	Plane 1.52 x 2.03 m, quad grid at about 12 mm spacing. Modifier order: Cloth, Solidify (15 mm, material offset 1 so the second face takes the silver material), Subdivision level 1. Slot 0 = quilted cotton, slot 1 = silver layer.
BLANKET_folded	Modelled, not simulated: the full blanket folded in thirds then in half, about 0.55 x 0.45 x 0.22 m, soft rounded fold edges, white side out.
SWATCH layers	Three separate planes, 0.50 x 0.50 m, each with a 2 mm solidify and one gently curled corner: SW_cotton (top), SW_fill (middle), SW_silver (bottom).
SNAP_stud	Metal press-stud, 12 mm across and 5 mm tall, with a 16 mm base ring. Sits 40 mm in from a corner of the silver face.
CORD	White round cable 3 mm thick built from a bezier curve, ending in a white moulded snap cap about 22 mm across with a metal socket underneath.
PLINTH	Cream block 1.2 x 0.8 x 0.9 m with a 6 mm bevel.
BEDROOM set	Queen bed with a low dark-oak frame and a white fitted mattress, cream wall behind, pale stone floor. No lamps, pictures or props that could carry text.

7. Materials

All materials use the Principled BSDF. Colours are starting points to be matched against the reference photos.

Material	Recipe
MAT_quilt_cotton	Base colour #F2F0EA, roughness 0.9, sheen weight 0.6, sheen roughness 0.5, subsurface weight 0.05. Quilting is true displacement, not bump: rotate object coordinates 45 degrees, scale so one diamond is 100 mm, and drive height with $ \sin(\pi \cdot u) \times \sin(\pi \cdot v) $ so each diamond puffs up and the stitch lines sit low. Height 9 mm. Add a fine weave bump from two crossed Wave textures at about 900 repeats per metre, strength 0.15. Set the material's displacement to 'Displacement and Bump'.
MAT_silver_layer	Base colour #8E9094, metallic 0.7, roughness 0.38, anisotropic 0.4. Weave bump from two crossed Wave textures at about 1200 repeats per metre, strength 0.25. Break up the roughness with a Noise texture (range 0.3 to 0.5) so individual threads glint as the camera moves. No grid lines: the page describes this face as smooth.
MAT_fill	Base colour #FAFAF7, roughness 1.0, sheen weight 1.0, subsurface weight 0.2, with a soft Noise displacement of 2 mm for a fibrous look.
MAT_steel	Base colour #C9CBCD, metallic 1.0, roughness 0.22.
MAT_cord_white	Base colour #F4F4F2, roughness 0.4, subsurface weight 0.1.
MAT_cream	Base colour #EFEAE0, roughness 0.7. Used for the plinth and walls.
MAT_oak_dark	Base colour #5A4A3C, roughness 0.5, with a stretched Noise texture for grain.
Pulse (added to MAT_silver_layer)	Emission masked to a soft ring that expands from the snap position. Mask = a 60 mm wide band at distance R from the snap, multiplied by the weave pattern so the light appears to run along the threads. Colour sampled from the logo file (expect a pale yellow near #F2EE8A). Emission strength peaks at 3. R is a custom property animated in shot 4.

8. Lighting

- **Key:** one large area light, 4 x 2.5 m, to camera left and slightly above, colour 5600 K, pointed across the product so the quilting casts soft shadows.
- **Window shadow:** a plane with four vertical slots in front of the key so soft bars of light fall across the wall, as in the brand's photos. Keep it subtle.
- **Fill:** world colour #F3EFE8 at strength 0.6. No HDRI is needed.
- **Rim for metal:** a narrow strip light behind and above in shots 3 and 4 so the silver layer and the snap catch a highlight.
- **Shot 5 dawn:** start the key at 40% strength and 4200 K and ramp to 100% and 5600 K across the shot.

9. Shot list

Each shot is its own scene with its own camera and a local frame range starting at 1. Positions are in metres as (x, y, z); the camera looks along +Y.

Shot	Global frames	Length	What happens
S1 Folded	1 to 90	3.0 s	The folded white blanket on the plinth. Slow push in.
S2 Unfurl	91 to 225	4.5 s	The blanket billows open in slow motion and turns, revealing the silver face.
S3 Exploded	226 to 375	5.0 s	A corner swatch separates into its layers while the camera orbits. Labels appear.
S4 Snap	376 to 465	3.0 s	Macro: the cord snaps onto the stud and a pale yellow pulse spreads through the threads.
S5 Bed	466 to 540	2.5 s	The blanket settles over a bed, white side up, as dawn light rises.
S6 Logo	541 to 600	2.0 s	Black card, logo, tagline.

S1 Folded (90 frames)

- Plinth top at $z = 0.9$. BLANKET_folded sits centred on it.
- Camera 50 mm, $f/4$, focus on the front fold. Moves from (0, -2.3, 1.30) to (0, -1.6, 1.18), aimed at (0, 0, 1.02). Linear interpolation, so the move never eases to a stop before the cut.
- No text.

S2 Unfurl (135 frames)

- Do not try to simulate the unfolding from the folded stack. Cut on motion: the shot opens with the blanket already in the air.
- BLANKET_full starts hanging by its two top corners at $z = 2.3$, white side to camera. Pin those corners to two empties. Over frames 1 to 60 the empties swing forward and up, then one crosses over the other so the cloth rolls and the silver face turns to camera by frame 90.
- Cloth: mass 0.45 kg, tension and compression 15, shear 5, bending 8, self-collision on at 6 mm, quality steps 12, speed multiplier 0.4 for the slow-motion feel. Bake to cache/s2 before rendering.
- Camera 35 mm, $f/5.6$, at (0, -3.6, 0.9) rising to (0, -3.4, 1.2), aimed at (0, 0, 1.4).
- Check at preview: by frame 100 at least half the frame is silver.

S3 Exploded (150 frames)

- The three swatch layers start stacked 2 mm apart at $z = 1.0$, tilted 15 degrees toward camera, with the snap on the silver layer's near corner.
- Frames 1 to 20: hold closed. Frames 20 to 80: layers separate to 130 mm gaps. Cotton starts moving at frame 20, fill at 28, silver at 36, each over 45 frames with a strong ease-out. Add a 3 degree counter-rotation between layers so they do not look like rigid cards.
- Camera 70 mm, $f/5.6$, orbiting 80 degrees around (0, 0, 1.0) at radius 1.5 m and height 1.25 m across the whole shot, constant speed.
- Labels (Sequencer text, thin 1 px leader lines), fading in over 12 frames: 'Quilted cotton exterior' at frame 85, 'Silver-thread conductive layer' at frame 100, 'Snap-button connection' at frame 115. The middle layer gets no label.

S4 Snap (90 frames)

- The silver face lies flat with soft wrinkles, stud near frame centre.
- Frames 1 to 24: the cord's snap cap lowers 40 mm onto the stud. Frame 25: contact, with a 1 mm overshoot and settle over 4 frames. This is the click.
- Frames 27 to 72: pulse radius R animates 0 to 0.6 m, emission 0 to 3 by frame 36 then back to 0 by frame 80.
- Camera 100 mm, $f/2.8$, focus on the stud, drifting from (0.10, -0.42, 0.22) to (0.06, -0.38, 0.20) relative to the stud.

S5 Bed (75 frames)

- BLANKET_full starts flat 0.35 m above the made bed, white side up, and is released at frame 1. Same cloth settings as S2 but speed multiplier 0.6. The bed and mattress are collision objects with 10 mm outer thickness.
- Camera 28 mm, $f/5.6$, from (0.9, -3.0, 1.5) pulling back to (1.1, -3.5, 1.7), aimed at the centre of the bed.
- Dawn light ramp as in section 8.

S6 Logo (60 frames)

- Background #0A0A0A (the brand's site colour). Logo on an emission plane at 42% of frame width, centred slightly above middle.

- Logo fades in over frames 1 to 15. Tagline 'Sleep grounded. Wake restored.' fades in over frames 15 to 30 beneath it, in the serif italic, off-white #F4F1EA.
- Optional small line at the bottom from frame 30, 60% opacity: 'Not intended to diagnose or treat any condition.'

Font: the brand typeface is not known. Use Playfair Display (regular and italic, free from Google Fonts) as a stand-in and swap it if the brand supplies its own.

10. Edit and sound

Assembled in a scene called EDIT using the Video Sequencer. All cuts are hard cuts except a 6-frame dip to black between S5 and S6.

Sound	Global frame	Note
Soft ambient music bed	1 to 600	Very quiet, no beat until S3. Fades out over the last 20 frames.
Fabric rustle and air	91 to 200	Follows the unfurl.
Three soft rising tones	246, 254, 262	One per layer as it lifts.
Metallic snap click	400	The loudest moment in the film. One clean click.
Low warm hum that resolves	402 to 541	Rises with the pulse, resolves into the logo card.

11. Build order and sign-off points

Claude Code must stop at each sign-off and show the stills before continuing. Full renders are the expensive step, so nothing is rendered in full until its still is approved.

Step	Work	Sign-off
1	Check Blender version and GPU, create folders, run 00_setup.py	Report version, device and one test render time
2	Build assets and materials. Render a material board: cotton, silver, fill, steel, cord side by side	Saurav approves materials against the reference photos
3	Build all six scenes with cameras and lights. One 48-sample still per shot at its key frame	Saurav approves framing and lighting
4	Animate and bake cloth. Viewport playblast of every shot at half resolution	Saurav approves motion and timing
5	Render one final-quality frame per shot and time it. Multiply by frame count for a total estimate	Saurav approves the time budget
6	Full render of all shots to PNG sequences	None
7	Sequencer edit with text and sound, export MP4	Final review

12. Kickoff prompt for Claude Code

Read Ellectrify_Blanket_Blender_Spec.pdf in this folder in full before doing anything.

Build the 20-second Ellectrify Grounding Blanket film exactly as specified, entirely in Blender, using Python scripts saved under scripts/.

Start with step 1 of section 11: find the Blender executable, report its version and confirm Cycles can use the RTX 4080 with OptiX. If the version is older than 4.2, tell me before continuing.

Then check assets/ against section 3 and list anything missing. Do not substitute or recreate the logo. If the logo is missing, build everything else and leave shot 6 with a placeholder card that says LOGO HERE.

Follow the build order in section 11 and stop at every sign-off point, saving stills to previews/ and telling me which files to look at.

Obey every rule in section 2. Never put text on a 3D object. Never add a label for the middle layer. Do not start a full render until I approve the per-shot stills and the time estimate.

13. Final checklist

- Exactly 600 frames, 1080 x 1920, 30 fps, with audio.
- The silver face is clearly visible by the midpoint of shot 2.
- Three labels only, spelled exactly as in section 9.
- No lettering on any object, no people, no spark or lightning effect.
- The pulse colour matches the logo yellow.
- The logo on the end card is the brand's own file, sharp and undistorted.
- No flicker or cloth passing through itself, the bed or the plinth.
- The click lands on the frame where the snap cap makes contact.

Sources: ellectrify.com product page for the Grounding Blanket, the Science and Compliance pages, and the brand's Instagram posts, read on 7 October 2026.